

Management of the margins in cutover bogs: ecological conditions and effects of afforestation

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Abstract Several studies contributed to the development of restoration techniques for open *Sphagnum*-dominated habitats on peat-extracted bogs. Yet, in exception to some afforestation efforts, connectivity between restored sites and surrounding landscapes has received little attention. The general goal of this study is to ameliorate management of very shallow peat fields (<30 cm) located within the margins of peat-extracted bogs. Firstly, to assist decision making in peatland management, baseline ecological conditions, peat physicochemistry and spontaneous vegetation recolonization were assessed for 18 of these fields. This first study revealed that (1) concentrations of several macro-nutrients are almost one order of magnitude lower for unrestored fields than previously characterized natural lagg habitats of the same region, and (2) there is little spontaneous colonization. In a second study, peat chemistry and soil/air microclimate were evaluated in plantations established on shallow residual peat (a 21-year-old *Larix laricina* plantation and an 18-year-old *Picea mariana* plantation) and compared to adjacent unrestored shallow bare-peat fields. This second study showed that afforested peat

fields are characterized by (1) a soil enriched in nutrients, notably in N, P, and K, and (2) a more humid and cooler microclimate at the soil/air interface, with less daily humidity and temperature fluctuations. These results indicate that afforestation is an appropriate approach to start an ecological recovery. Yet, the absence of natural recolonization by herbs and mosses in the understory of afforested peat fields suggests that reintroduction of appropriate understory species should also be considered if the goal is to restore a fully functional ecosystem.

Keywords Cutaway · Industrial · Lagg restoration · Peat extraction · Post-extraction

Introduction

The understanding of undesirable alternate stable-state ecosystems is one of the major challenges in restoration ecology (Bachmann et al. 1999; Seastedt et al. 2008). The study of their ecological filters and associated positive feedback loops often provides insights for the development of new restoration techniques. For example, studies in cutover bogs (peat-extracted ombrotrophic peatlands) have shown that, following the cessation of peat extraction activities, recolonization by bog species is impeded by several ecological filters, such as limited long-distance dispersal of propagules (Campbell et al. 2003), severe

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droughts and hot temperatures due to low albedo of substrate (Price et al. 1998), and strong frost-heaving in spring (Groeneveld and Rochefort 2005). Using this knowledge on ecological filters, peatland ecologists have developed techniques to restore the open expanse of *Sphagnum* bog habitat on the dome of cutover bogs, i.e. in peat-extracted areas with substantial amount of peat remaining (Rochefort and Lode 2006; Graf and Rochefort 2016).

Restoration of shallow-peat areas has however been lagging behind (Grand-Clement et al. 2015). In the context of ecological restoration of cutover bogs, these areas, hereafter called cutaway fields, are most often located at the edges of the peat massif, in the former lagg zone (i.e. the minerotrophic depression in the margin of a bog; the bog-mineral land ecotone). As wetland hydrology is a major feature defining lags, restoring a functioning lagg will inevitably require rewetting (Howie and Tromp-van Meerveld 2011). Interestingly, in the last few years, several projects, located mostly in the UK, aimed at adapting rewetting methods for shallow-peat areas (Grand-Clement et al. 2015). Yet, in practice, definition of “realistic restoration goals” (Hobbs 2007) for cutaway fields located in the margins of a peat massif is often highly influenced by social and economic considerations. Because these areas are often abandoned first during the commercial life of an excavated peat deposit, rewetting is incompatible with peat extraction activities still occurring in adjacent areas. Restoration of a functioning lagg is thus postpone after complete abandonment of peat extraction activities in central areas of the dome, which usually occurs years to decades later than the cessation of activities on the shallower marginal areas (personal observations). As a result, the few studies which have addressed the rehabilitation of these areas in North-America (Bussi eres et al. 2008; Caisse et al. 2008) have only focused on adapting European afforestation techniques (reviewed in Paavilainen and P  iv  nen 1995). Afforestation, through the planting of conifer seedlings (most often *Larix laricina* or *Picea mariana*), is now the dominant practice adopted by peat industry to revegetate marginal areas of post-extracted peatlands of eastern North-America (Hugron et al. 2013).

Current North-American and European afforestation practices offer two major advantages: (1) trees can revegetate sites rapidly, when provided with an appropriate initial fertilization, and (2) afforestation

is compatible with the maintenance of an active drainage system, required for the continuation of adjacent peat extraction activities if still occurring (Paavilainen and P  iv  nen 1995; Aro and Kaunisto 1998; Renou and Farrell 2005; Bussi eres et al. 2008; Renou-Wilson et al. 2008, 2010). Yet, major questions remain on the usefulness of the current afforestation practices. What are the potential ecosystem engineering effects of planted trees? Trees can further lower water table level by evapotranspiration (Sun et al. 2001), but also improve understory microclimate in dry conditions by reducing solar radiations through a developing canopy (Kellman and Kading 1992; Holmgren et al. 1997; Hart and Chen 2006). On the aspect of soil chemistry, trees can increase some nutrient pools in soils through a nutrient uplift mechanism (Jobb  gy and Jackson 2004), but trees can also deplete soils of some nutrients through the sequestration of nutrients in biomass (Berthrong et al. 2009). Consequently, do trees improve or deteriorate the ecological conditions of cutaway fields?

Another aspect which has been neglected in the management of these sites is the return of an appropriate lagg vegetation community. In natural peatlands, the lagg habitat is characterized by a unique assemblage of ecotonal plant species, of which many are restricted to the more minerotrophic wetland conditions, and not typically found in adjacent bog and mineral land (Paradis et al. 2015). For example, in eastern Canada, tall shrub species, such as *Alnus incana* ssp. *rugosa*, *Ilex mucronata*, *Viburnum nudum* ssp. *cassinoides*, usually form the dominant stratum in the lagg habitat, although coniferous trees are also often present (Paradis et al. 2015). Does spontaneous recolonization by these lagg species occur on cutaway fields? If so, how does recolonization differ between unrestored cutaway fields and afforested cutaway fields? Do tree plantations follow an “ecological trajectory” conducive to the restoration of a lagg-swamp ecosystem?

In this paper, a general goal is to ameliorate management of cutaway fields located in margins of cutover bogs using the ecosystem restoration framework developed by Graf and Rochefort (2016). A first objective is to increase our knowledge on baseline ecological conditions of these peat extraction fields, through a study of peat physicochemistry and spontaneous vegetation recolonization. Second objective is to assess effects of afforestation through a comparison

of ecological conditions of afforested and unrestored cutaway fields; focus is on soil chemistry and soil/air microclimate, two factors that often limit success of reintroduced plants during restoration (Rocheffort and Bastien 1998; Groeneveld and Rocheffort 2005; Caisse et al. 2008; González et al. 2013). Finally, after these two assessments, it will be possible to define management interventions required, if any, for cutaway fields and afforested peat fields in order to re-establish an adequate ecological trajectory towards natural reference lagg-swamp ecosystems.

Methodology

Study sites

All peatlands included in this study (Table 1; Fig. 1) are located in the “Eastern Lowlands Ecoregion” of New Brunswick (eastern Canada), a region characterized by a low relief, poor soil drainage and high soil acidity (TECWG 2007). In New Brunswick, almost all peat extraction activities occur in this ecoregion. According to a local weather station, annual total precipitation was approximately 1171 mm during the

1992–2011 decade (weather station 8100503, datas from Environment Canada 2012).

In the preparation of natural peatlands for extraction through vacuum-milling, a technique highly dependent on the drying of peat by sun and wind, peatlands are broken up into smaller peat extraction field areas, delimited by drainage ditches. It is these peat extraction fields that form the standard characterization unit used throughout this study. Unrestored cutaway peat extraction fields were visited in marginal areas of ten cutover bogs which have been extracted using mechanical vacuum-milling technique. As characterization was intended for the amelioration of management of the shallow margins of active peat-extracting peatlands, fields were inspected in order to find representative areas that share these properties: (1) there is a very shallow residual peat (between 10 and 30 cm), (2) peat extraction activities have ceased at least five years prior to sampling, but a functional drainage system is still present (peat extraction still occurring in adjacent fields). After visiting all peatlands, 18 fields were characterized inside the seven peatlands shown on Fig. 1 ((3) out of the 10 visited peatlands were discarded because they did not meet all criteria). In a given peatland, the two or three

Table 1 Locations of all sampled cutaway fields (1a to 7b) and the two coniferous plantations (7c, d)

Cutaway fields and plantations	Latitude	Longitude	Nearest town
1a	47°42'36"N	64°50'23"W	Inkerman-ferry
1b	47°42'31"N	64°50'20"W	Inkerman-ferry
2a	47°37'35"N	64°50'26"W	Inkerman
2b	47°37'41"N	64°48'28"W	Inkerman
3a	47°50'16"N	64°34'38"W	Coteau-road
3b	47°50'58"N	64°33'13"W	Coteau-road
3c	47°50'37"N	64°34'45"W	Coteau-road
4a	47°49'24"N	64°38'16"W	Lameque-portage (site 1)
4b	47°49'24"N	64°38'17"W	Lameque-portage (site 1)
4c	47°49'46"N	64°37'44"W	Lameque-portage (site 1)
5a	47°49'47"N	64°38'34"W	Lameque-portage (site 2)
5b	47°49'57"N	64°38'46"W	Lameque-portage (site 2)
5c	47°49'56"N	64°38'32"W	Lameque-portage (site 2)
6a	47°47'37"N	64°35'13"W	Saint-raphael
6b	47°47'39"N	64°35'23"W	Saint-raphael
6c	47°47'40"N	64°35'34"W	Saint-raphael
7a	46°57'46"N	64°51'44"W	Pointe-sapin
7b	46°57'45"N	64°51'45"W	Pointe-sapin
7c <i>Larix</i>	46°57'48"N	64°51'36"W	Pointe-sapin
7d <i>Picea</i>	46°57'45"N	64°51'32"W	Pointe-sapin

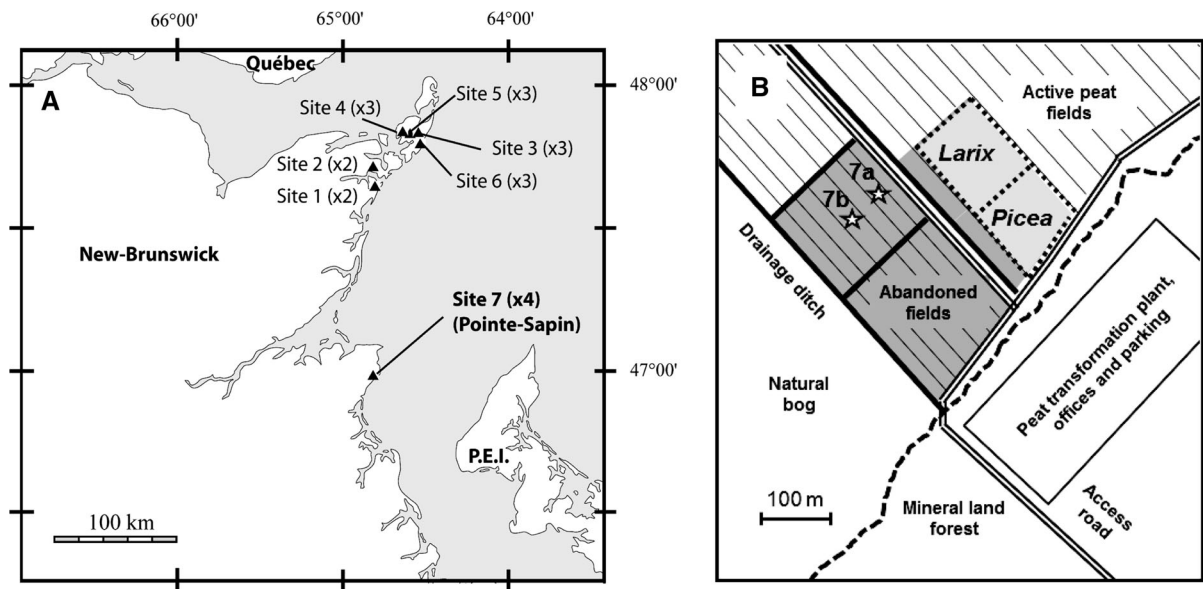


Fig. 1 **a** Location of all sampled peatlands (with number of sampled cutaway fields in parenthesis). **b** Location of the four sampled sites (7a, b, *Larix*, *Picea*) inside Pointe-Sapin peatland

characterized fields were chosen as the most representative subset of the abandoned area; characterized fields were never directly adjacent, but spaced at least by one peat field.

Site selected for characterization of the effects of afforestation (Table 1; Figs. 1, 2) is a peatland sector of 2.5 hectares (approximately 100 × 250 m), in which, following cessation of peat extraction activities, one half of the area has been planted with black spruce (*P. mariana*) in 1991 and other half with eastern

larch (*L. laricina*) in 1994 (Bussi res et al. 2008). The site offers an excellent opportunity to study possible long-term effects of tree plantation on ecosystem attributes for two reasons: (1) these two coniferous plantations on cutaway fields are among the oldest ones in North America; (2) compared to the few other plantations 15 to 20-year-old in the region, the chosen peatland was ideal for our purpose as adjacent bare-peat extraction fields were left unrestored (no intervention since abandonment) to allow comparison.



Fig. 2 **a** Picture showing a typical cutaway area with spontaneous recolonization restricted to the inside of ditches. **b** Picture showing the dense litter in the understory in *Larix* plantation

Methods

Spontaneous vegetation recolonization

All cutaway fields were inspected for spontaneous plant recolonization in August–September 2011, late summer being the best period for the identification of helophyte species (light species). At every 30 cm all along the ditch, width of the vegetation band associated with the ditch micro-ecosystem was measured as the point where plants became more sparsely distributed, i.e. where spacing between individual plants exceeded 0.5 m. Vegetation band was delimited in three zones: center of peat extracted area, vegetated band on margins of ditches and inside ditches. Vegetation abundance cover was evaluated in each zone separately.

Soil physicochemistry

Three surficial peat soil samples (from the top 5–7 cm excluding surficial soil crust) were taken in the center of each cutaway field; samples were then mixed together prior to analysis. Three surficial soil samples (from coniferous litter) were also taken in each plantation, and mixed together prior to analysis. Bulk density was calculated as the oven-dry (105 °C) mass of a known volume of peat (Hillel 1998). Soils samples were also analyzed for pH, EC, and nutrients (total nitrogen content, total phosphorus content, extractable concentrations in potassium, calcium, magnesium, manganese, sodium, and iron) following the protocol described in Paradis et al. (2015). In addition, available phosphorus (PBray) was extracted following standard extraction procedure (Bray and Kurtz 1945) and concentrations were measured through an ICP Optima4300DV (Perkin-Elmer); available ammonium was extracted following standard extraction procedure (Quikchem method 10-107-06-2-B), and concentrations measured on a FIA Quikchem 4000 (Lachat Instruments).

One method increasingly used to evaluate the relative ecological state of an ecosystem is to use ordinations, where groups of sites can be compared through evaluation of ecological distances that combine numerous variables (Graf et al. 2008; González et al. 2013). In our case, Principle Component Analysis (PCA), a relatively simple ordination technique for multivariate data (Legendre and Legendre

2012), was used to compare soil physicochemical properties of the following three ecological groups: (1) cutaway peat fields (current study), (2) afforested peat fields (current study), and (3) reference natural lagg-swamps from Paradis et al. (2015).

Light environment and soil/air microclimate

To interpret results from microclimate characterization, canopy cover was characterized inside plantations using hemispherical photography. Four pictures were taken in each plantation on three dates (20-April; 7-August; 7-September) during the 2012 growing season (total of 24 pictures) and images were analyzed using Gap Light Analyzer software version 2.0 (Cary Institute of Ecosystem Studies).

Effects of afforestation on understory microclimate (soil/air humidity and temperature) were assessed using HOBO U23 Pro v2 Temperature/Relative Humidity Data Logger (Onset Computer Corporation); two sensors were installed at the soil/air interface in each plantation and in one adjacent unrestored cutaway field, for a total of six sensors. At each visit to the peatland (three times during growing season 2011 and three times in 2012), level of water table was also measured in the two plantations and the adjacent unrestored field.

Results

Spontaneous vegetation recolonization

Vegetation bands were successfully delimited in all cutaway fields. The relatively bare peat extracted area was clearly the most important band in each cutaway field (approximately 20 m wide out of the usually 25–30 m peat fields). The plant cover in these large bare peat bands was always less than 2 % and, usually, only a few clumps of *Eriophorum vaginatum* ssp. *spissum* were present sparsely (Fig. 2a). Spontaneous revegetation was particularly abundant in a narrow zone of influence nearby drainage ditches (Table 2). In fact, on the margins of ditches, vegetation was always restricted to a narrow band where peat was mixed with some mineral soil from the ditch, a band in average 1.4 m wide (± 0.5 m); plant community was dominated by ruderal species, most often by two or three species among *Bidens frondosa*, *Agrostis alba/scabra*,

Table 2 Dominant vegetation inside ditches and on the margin of ditches of unrestored cutaway fields

Site	Ditch margins											
	<i>Agrostis alba</i>	<i>Anaphalis margaritaceae</i>	<i>Betula papyrifera</i>	<i>Bidens frondosa</i>	<i>Epilobium angustifolium</i>	<i>Euthamia graminifolia</i>	<i>Juncus effusus</i>	<i>Picea mariana</i>	<i>Polytrichum sp.</i>	<i>Rumex acetosella</i>	<i>Scirpus cyperinus</i>	
1a											x	
1b	x			x								
2a	x			x		x						
2b	x			x			x					
3a				x		x				x		
3b	x						x			x		
3c	x						x			x		
4a			x					x	x			
4b			x									
4c		x			x							
5a		x				x						
5b	x			x								
5c	x			x			x					
6a	x			x			x					
6b	x			x								
6c	x			x		x						
7a											x	
7b												
7c											x	
7d												

Site	Center of ditches								
	<i>Agrostis alba</i>	<i>Bidens frondosa</i>	<i>Carex canescens</i>	<i>Juncus effusus</i>	<i>Polytrichum sp.</i>	<i>Rhododendron canadense</i>	<i>Rumex acetosella</i>	<i>Scirpus cyperinus</i>	<i>Typha latifolia</i>
1a		x			x			x	
1b					x			x	
2a									
2b					x				
3a	x			x			x		
3b	x			x			x		
3c	x			x			x		

Table 2 continued

Site	Center of ditches									
	<i>Agrostis alba</i>	<i>Bidens frondosa</i>	<i>Carex canescens</i>	<i>Juncus effusus</i>	<i>Polytrichum sp.</i>	<i>Rhododendron canadense</i>	<i>Rumex acetosella</i>	<i>Scirpus cyperinus</i>	<i>Typha latifolia</i>	
4a			x		x	x				
4b			x		x	x				
4c					x			x		
5a					x			x	x	
5b										
5c					x					
6a								x		
6b										
6c										
7a					x			x		
7b					x			x		
7c			x							
7d			x							

Juncus brevicaudatus, *Rumex acetosella*, *Euthamia (Solidago) graminifolia*. Inside ditches, aquatic macrophytes (mainly *Scirpus cyperinus*) were the dominant species. At the interface between the two communities, on the vertical wall of the ditches, ground layer was usually dominated by *Polytrichum strictum*, often intermixed with other small erect mosses such as *Dicranella cerviculata*, *Ceratodon purpureus*, and *Pohlia nutans*.

Similarly to unrestored cutaway fields, very little recolonization took place in coniferous plantations. Total cover was less than 1 % (aside from the trees). Under the trees, on the former peat-extracted surface, only small mosses occurred very sparsely. Inside the former ditches, which are almost filled up with conifer needles, small clumps of *Carex canescens* also occurred sparsely (Fig. 2b).

Soil physicochemistry

Soils from the two main groups, cutaway fields and natural lagg-swamps, clearly differ on several aspects as there is no overlap in the confidence ellipses on the ordination shown on Fig. 3. In this ordination, first axis (PC1), which explains 36 % of the variance, is more related to the variables Total N and Ca. Second

axis (PC2), which explains 18 % of the variance, is more related to the variables Mg, Na and PBray.

The new coniferous humus soils differ from residual peat of adjacent unrestored cutaway fields on several aspects (Table 3). First, bulk density is lower (0.07 in plantations vs 0.15 in unrestored fields; $P < 0.05$), which is due to the very coarse nature of the litter, constituted of conifer needles and small branches. Interestingly, these new soils are particularly rich in phosphorus and potassium, two major macronutrients for plants. Total phosphorus concentration is 2.1 times higher in *Larix* plantation and 1.7 times higher in *Picea* plantation than in nearest unrestored field ($P < 0.05$). For potassium, exchangeable concentrations are respectively 13 times higher in *Larix* plantation, and 11 times higher in *Picea* plantation ($P < 0.05$). Furthermore, total P and exchangeable K values measured in soils of plantations are not only higher than adjacent control unrestored fields, but they do not fall in the range of values of the 18 unrestored sites (1.83 and 1.44 mg/g total P in *Larix* and *Picea* plantations vs 0.50 ± 0.33 in unrestored fields; 1.17 and 1.01 mg/g exchangeable K in *Larix* and *Picea* plantations vs 0.10 ± 0.07 in unrestored fields). Calcium and manganese are two other cations which are also particularly more

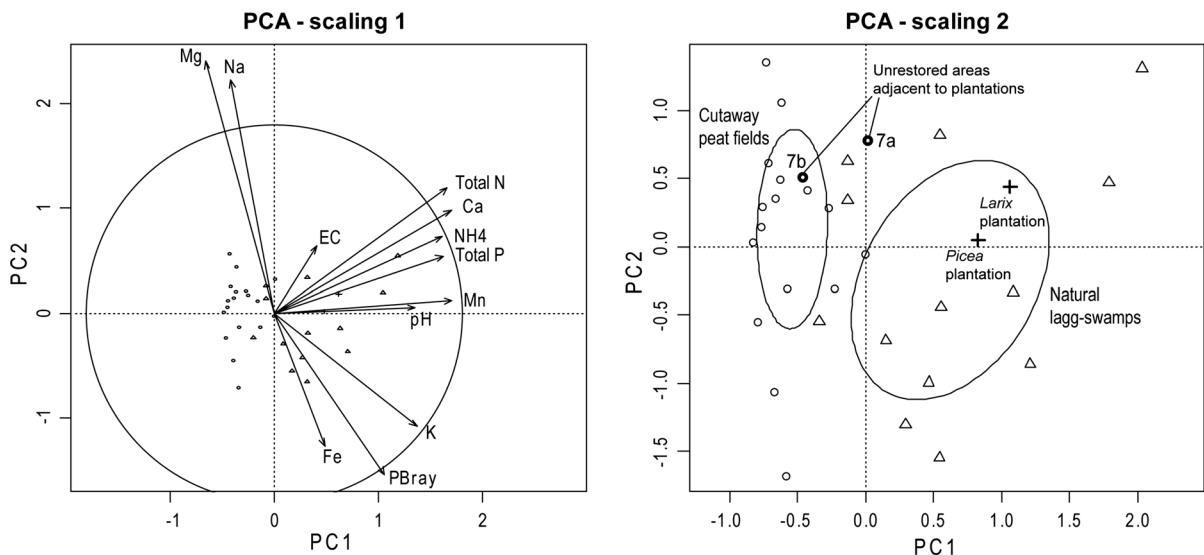


Fig. 3 PCA comparing soil chemistry data from natural lagg-swamps (Paradis et al. 2015) and data from current study (plantations and unrestored areas) (analysis carried using R; R Development Core Team 2008). Dataset for PCA using (standardized) soil chemistry variables. Scaling 1 emphasizes

relation between soil variables, where *arrows* represent different soil variables. Scaling 2 emphasizes relation between sites. First axis (PC1) explains 36 % of the variance, while second axis (PC2) explains 18 %

Table 3 Peat soil chemistry in all sampled cutaway fields, and inside the two coniferous plantations

Site type	Site id	Bulk density (mg/L)	pH	EC (μS/cm)	N_tot (mg/g)	NH4 (mg/g)	P_tot (mg/g)	PBray (mg/g)	K (mg/g)	Ca (mg/g)	Mg (mg/g)	Mn (mg/g)	Na (mg/g)	Fe (mg/g)
Cutaway fields	1a	0.15	3.8	10	6.3	0.042	0.83	0.001	0.11	0.84	0.95	0.009	0.59	0.068
	1b	0.15	3.6	0	2.8	0.012	0.40	0.161	0.04	0.14	0.15	0.001	0.03	0.033
	2a	0.27	3.7	30	3.9	0.013	0.38	0.080	0.05	0.36	0.29	0.005	0.14	0.075
	2b	0.11	3.7	30	8.2	0.035	0.31	0.007	0.11	1.02	1.04	0.037	0.37	0.104
	3a	0.11	3.5	0	8.0	0.060	0.92	0.011	0.22	1.35	1.13	0.023	0.73	0.054
	3b	0.09	3.5	0	7.0	0.047	0.87	0.005	0.30	1.32	1.52	0.027	0.71	0.024
	3c	0.12	3.9	32	10.7	0.086	0.50	0.014	0.16	0.81	1.05	0.049	0.41	0.176
	4a	0.14	3.7	34	9.2	0.028	0.19	0.010	0.05	0.57	0.96	0.016	0.28	0.071
	4b	0.14	3.7	19	6.0	0.034	0.13	0.003	0.04	0.70	0.85	0.024	0.31	0.076
	4c	0.15	3.5	47	9.5	0.104	0.37	0.007	0.08	0.94	1.12	0.016	0.32	0.108
	5a	0.12	4.2	77	13.9	0.122	0.33	0.010	0.11	2.19	0.53	0.043	0.18	0.126
	5b	0.15	3.8	25	9.1	0.065	0.34	0.010	0.08	0.54	0.71	0.022	0.17	0.168
	5c	0.12	3.8	18	13.7	0.171	1.04	0.003	0.10	0.49	0.91	0.010	0.26	0.166
	6a	0.33	4.1	122	1.7	0.031	0.15	0.008	0.04	0.39	0.33	0.010	0.27	0.039
	6b	0.15	3.9	49	5.2	0.068	0.11	0.021	0.07	0.85	1.17	0.024	0.36	0.131
	6c	0.17	3.7	29	8.0	0.027	0.17	0.148	0.09	2.61	0.63	0.030	0.40	0.072
	7a	0.12	3.4	0	11.3	0.424	0.86	0.007	0.09	1.55	0.66	0.025	0.54	0.110
	7b	0.13	3.5	64	11.4	0.049	1.04	0.007	0.07	0.91	0.55	0.013	0.60	0.096
	Mean	0.15	3.7	33	8.1	0.079	0.50	0.028	0.10	0.98	0.81	0.021	0.37	0.094
	StdDev	0.06	0.2	31	3.4	0.095	0.33	0.049	0.07	0.64	0.36	0.013	0.20	0.046
Plantations	7c <i>Larix</i>	0.07	3.8	326	15.8	0.091	1.83	0.109	1.17	3.44	1.05	0.136	0.13	0.020
	7d <i>Picea</i>	0.07	3.6	241	9.9	0.195	1.44	0.092	1.01	2.05	0.84	0.193	0.13	0.023
	Mean	0.07	3.7	284	12.9	0.143	1.64	0.101	1.09	2.75	0.95	0.165	0.13	0.022
	StdDev	0.02	0.6	68	6.8	0.212	0.48	0.091	1.42	1.28	0.26	0.140	0.21	0.220

For easier comparisons, the average values taken from a study on natural lags (Paradis et al. 2015) were added

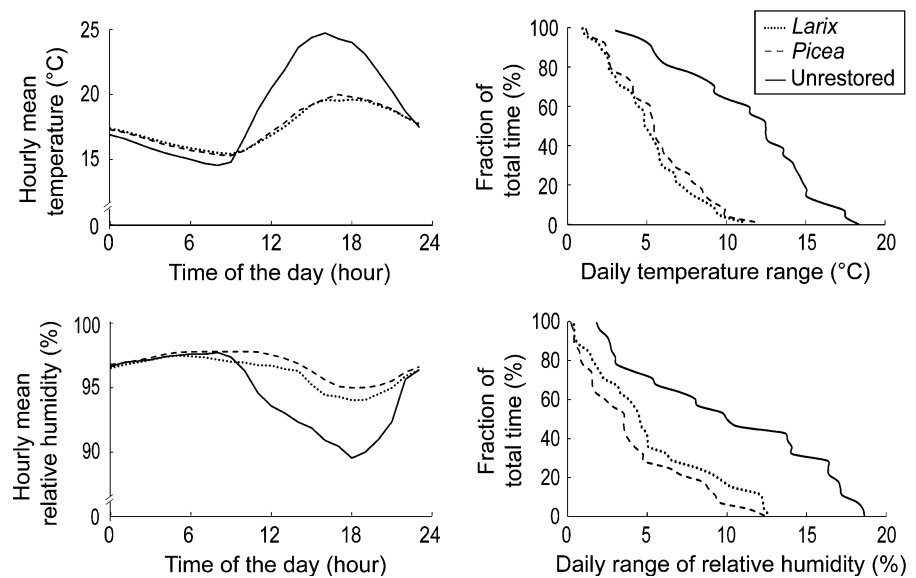
abundant in the new soils compared to unrestored fields. pH is not significantly different.

Light environment and soil/air microclimate

In *Picea* plantation, shading provided by the trees stayed similar during the different period of the growing season (84 ± 4 % in spring, 88 ± 7 % in mid-summer, 88 ± 5 % in late summer). In contrast, it fluctuated widely in *Larix* plantation due to the deciduous habit of this species (36 ± 2 % before leaf emergence, 85 ± 3 % in mid-summer, 76 ± 5 % in late summer).

Measured microclimate environment variables were very similar for the two coniferous species during the studied period (August 2011). Compared to adjacent unrestored cutaway fields, both temperature and humidity were more stable inside the two plantations (Fig. 4). Maximal temperatures and average temperatures were strongly reduced inside plantations, while average substrate humidity was higher (Fig. 4). Water table usually stayed below 70 cm inside plantations (only exception in spring 2012, where ice was found at approximately 50 cm below surface), but it fluctuated in the 20–70 cm below surface in adjacent unrestored fields (personal observations).

Fig. 4 Microclimate at the soil/air interface inside two coniferous plantations and one adjacent unrestored cutaway field (Pointe-Sapin, New Brunswick) during the month of August 2011 (672 hourly measurements from 4-Aug-2011 00:00 to 31-Aug-2011 23:00 in each site)



Discussion

Knowledge of baseline conditions: comparison of unrestored cutaway peat fields and natural reference ecosystems

In a recent paper on the bog-lagg-mineral land gradient in eastern Canada (Paradis et al. 2015), several ecological aspects of natural lagg-swamps were characterized with a methodology similar to the one used in this study. So, how do the environmental conditions of cutaway fields differ from the natural lagg-swamps? On the aspect of peat physicochemistry, several variables show that residual peat is poorer in nutrients: (1) total N content is about half (8 in cutaway fields vs 16 mg/g in natural lagg-swamps), (2) total P content is 3.7 times lower (0.5 in cutaway fields vs 1.8 mg/g in natural lagg-swamps), (3) concentrations in exchangeable base cations Ca, K, Mg, Mn are significantly lower than in natural lagg-swamps (Table 3). In addition, it is important to highlight that pH is significantly lower: 3.7 in unrestored fields versus 4.4 in natural lagg-swamps; because pH is closely related to bioavailability for many nutrients, the lower pH in cutaway fields also indicates a lower bioavailability (Chapin III 1980). Considering the close contact to mineral soil (less than 30 cm residual peat), the low concentrations in base cations may at first seem surprising. Yet, presence of an active drainage system considerably limits possible

enrichment from flooding event (snowmelt, rainy storm). Also the presence of a layer of peat is known to impede the transport of cations, particularly if the soil surface is relatively bare of vegetation (Rezanezhad et al. 2012).

The spatial distribution and species composition of natural recolonization, which was strictly restricted to ditches (less than 2 % of vegetation cover on abandoned areas), suggest that active drainage system and absence of mineral enrichment may play important roles as limiting factors in the recovery of the system. Also, it is interesting to note that almost all dominant species that naturally recolonized ditches are classified as helophyte according to local literature (Hinds 2000), and none can be considered typical lagg-swamp species (Paradis et al. 2015). Overall, our findings resembles more to those described in extracted peatlands with ombrotrophic residual peat (Wind-Mulder et al. 1996; Andersen et al. 2011) than in extracted peatlands with minerotrophic residual peat (Graf et al. 2008): cutaway fields do seem arrested in a stable state equilibrium.

Do conifer tree plantations follow an “ecological trajectory” conducive to the restoration of a lagg-swamp ecosystem?

In contrast to unrestored cutaway fields, plantations do seem to evolve on a right ecological trajectory following afforestation, at least for soil

physicochemistry and soil/air microclimate. Firstly, for soil physicochemistry, the developing soils tend to be relatively more similar to those of natural lagg-swamps than to the residual peat substrates found post peat extraction activities (Fig. 3). In fact, on the PCA ordination, the two points associated with soils of plantations fall inside the confidence ellipse for natural lagg-swamps (which encompasses 50 % of the points). It is difficult to really pinpoint the origin of nutrients in the soils of plantations: an initial fertilization was done at plantation time (Bussi res et al. 2008), but the sampled superficial humus layer was formed following plantation. Therefore, nutrients probably originate from both a recycling of first fertilization, and from underlying mineral layer. Indeed, various plant-mediated phenomena, such as the nutrient uplift mechanism (Jobb gy and Jackson 2004) and the increase in mineral weathering by roots and rhizosphere (Kelly et al. 1998), are already known to contribute in soil formation worldwide. For peat-extracted peatlands, many efforts have been given to understand effects of nutrient limitations on tree growth in cutaway and cutover areas (Bussi res et al. 2008; Caisse et al. 2008; Renou and Farrell 2005), but further studies are definitely needed to really understand long-term effects of afforestation on soil development.

In addition to amelioration of soil physicochemistry, study of soil/air microclimate has shown that afforested fields are characterized with a milder and more stable microclimate. These findings on effects of afforestation on microclimate concur with results from similar studies in other landscape settings, which highlight that reduction in solar radiation is a major cause for the observed differences in soil/air microclimate in forests (reviewed in Aussenac 2000).

As this study focused on cutover areas adjacent to active peat extraction areas, hydrological characterization was not a major goal of this study. Yet, it is interesting to note a few points about our observations on water table levels: (1) considering that summer 2011 was relatively humid (total precipitations of 233 mm during the studied time period of July–August 2011, compared to the average 164 mm of the 1992–2011 time period, Bas Caraquet stations; Environment Canada 2012), water table probably never reach higher level in average years and it is likely that the measured ameliorations in microclimatic conditions are more important in average years; (2) water

table levels are definitely not those typical of natural lagg-swamps, where water table levels usually stays near surface during growing period (Paradis et al. 2015). Although it is well known that water table level influences various peat chemistry variables, notably through some effects on redox potentials and microbial activity, we do not believe that any of the differences observed between the different ecosystems can be attributed to the abnormally wet conditions of 2011 as soils from unrestored and afforested fields were collected at same time.

Our few observation on water table levels are consistent with the concept of an increase in “biological drainage” caused by an increase in evapotranspiration following peatland afforestation (Jutras et al. 2006), with water table lower in plantations than in unrestored fields. More studies on the linkage between afforestation and its effects on water table and microclimate are however needed, as contrasting results were described in a recent study on the effect of birch afforestation on water loss in peatland (Limpens et al. 2014). Further studies could also focus on tree density and spatial effects of afforested areas on water loss; a study in natural mires has found that trees could produce higher evapotranspiration loss when clustered as tree isles compared to coherent forest expansion (Grygoruk et al. 2014).

Interestingly, amelioration in soil physicochemistry and soil microclimate did not result in an improved plant recolonization. In fact, only a few shade tolerant species were found sparsely under plantations. Absence of helophyte species, typically found in ditches of cutaway fields, is probably attributed to the dense shade created by overstory. In contrast, absence of lagg-swamp species is probably attributed to a combination of dispersal constraints and lack of proper hydrological conditions.

Conclusion—implications for management

Results from our study suggest that evolution of both unrestored and afforested peat fields are arrested into alternate stable state equilibriums. We suspect that presence of active drainage ditches limits to some extent the natural recovery of these sites, and we agree with other authors that restoration of a functioning lagg ecosystem with minerotrophic conditions is tightly linked to restoration of a proper hydrology,

which can be done through sequential blocking of drainage ditch system or spatially adapted (Howie and Tromp-van Meerveld 2011; Malloy and Price 2014). Yet, careful monitoring is also needed to better understand possible internal dynamics establishing in these novel ecosystems (Graf and Rochefort 2016). In particular, we highlight two aspects that should be monitored following rewetting in order to improve restoration strategies. Firstly, effect of rewetting on survival of trees is one important question that must be addressed. According to existing literature on forested wetland restoration (mostly on former agricultural lands), it is usually agreed that successful restoration requires three key management actions: (1) creation of microtopography, (2) plantation of woody species on mounds, and (3) rewetting (Whitelaw et al. 1989; Barry et al. 1996; Buck 2003; Allen et al. 2004; Simmons et al. 2011; Kangas 2012). Although presence of aerated mounds is essential for tree survival during the periodic flooding that naturally occurs in natural swamps (Titus 1990; Ehrenfeld 1995; Chimner and Hart 1996), very little attention has been given to restoration of microtopography in peatland afforestation trials. Both black spruce and tamarack are tolerant to high water table levels (Lieffers and Rothwell 1986; MacDonald and Lieffers 1990), but total absence of mound-and-pit microtopography in afforested cut-away fields can have serious implications on the potential effects of rewetting on planted trees. In fact, in the course of forested wetland restoration, extreme meteorological events can create amplified flooding conditions, and role of microtopography can be even more important in increasing survival of flood intolerant species (Simmons et al. 2011). Personal observations in a young plantation flooded by a beaver dam (48°29'26.6496"N, 68°27'33.2856"W) indicate that death of planted trees is a possible outcome from rewetting if flooding last for a long period annually.

Secondly, effect of rewetting on natural plant recolonization should also be monitored very closely, and management should be adapted in order to take into account presence of potentially invasive species that could cause a switch to an unwanted stable-state ecosystem. In fact, two species already present inside ditches pose potential threats. *S. cyperinus*, a fen species, is a good colonizer of extracted peatlands with residual fen peat, but it is sometimes seen as a weedy species

(Graf et al. 2008; MacKay et al. 2010). *Typha latifolia*, a marsh species, is already known to show invasive behavior in restored peatlands, especially along pool margins and along former ditches (Fontaine et al. 2007). Once well established, these two tall macrophytes can dominate vegetation for relatively long periods: after 20 years, the two species were still dominant (total cover of 59 %) in two created depressional wetlands of Virginia (Atkinson et al. 2005). As both species are totally absent from natural lagg habitats (Paradis et al. 2015), management should be adapted to limit further colonization by these species in restored peatlands.

In order to ameliorate ecosystem resistance, one suggested technique is to actively reintroduce lagg species for which natural recolonization may be partly limited by dispersal constraints. Notably, establishing shrubs, which are usually well adapted to flooding (Golet et al. 1993), and dominant inside the natural lagg habitat (Paradis et al. 2015), could (1) increase chances that sites stay vegetated following rewetting, and (2) create a denser canopy and control invasion by opportunistic species. Control through shading by trees is a technique that has already been applied for various helophyte species, such as *Calamagrostis canadensis* (Lieffers and Macdonald 1993) and *Phalaris arundinaceae* (Hovick and Reinartz 2007). In addition to shrubs, restoring a carpet of bryophytes could also contribute to lower risks of invasion by *Typha* and *Scirpus*, since these helophyte species do not germinate as well inside dense bryophyte carpets (Bourgeois et al. 2012; Lajoie 2015). In addition to an increased ecosystem resistance to stressors, reintroduction of ecotonal species would also ameliorate intrinsic wildlife habitat value of marginal cutaway fields, and increase the landscape structural connectivity between restored bog and mineral lands (Bell et al. 1997; Ehrenfeld and Toth 1997).

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